



Date	XXXXXX
Team ID	XXXXXX
Project Name	Virtual Agent-Driven SLA Breach Awareness & Justification System

Virtual Agent-Driven SLA System Data Flow

This document outlines the movement of data between external entities, automated processes, and data stores within the ServiceNow SLA governance framework.

System Component Legend

Using the standard DFD symbol logic, the following table defines the core architectural elements of the system.

Component Type	Name	Description
External Entity	Assigned To (ITIL User)	The persona who receives notifications and provides breach justification.
Process 1	Flow Designer	Triggers "SLA at Risk" notifications when the business elapsed percentage $\geq$ 80%.

Component Type	Name	Description
Process 2	Virtual Agent	Conversational interface that captures user acknowledgement and justification.
Data Store 1	Task SLA Table	Holds the running SLA records, durations, and percentage data.
Data Store 2	Incident Table	Primary record where justification, reason codes, and risk flags are stored.
Notification	Email Engine	Sends immediate action alerts to the Assignee and their Manager.

Operational Data Flow Sequence

The following sequence describes how data transforms from an incident creation to executive reporting.

1. **SLA Attachment:** An Incident is created, triggering the "Incident Resolution - Virtual Agent Governance" SLA definition with an 8-hour duration.
2. **Risk Monitoring:** The system monitors the **Task SLA** table. When the elapsed percentage hits 80%, the **Flow Designer** process activates.
3. **Governance Activation:**
 - The "SLA at Risk" flag is set to **True** on the Incident record.
 - Work notes are automatically updated: "SLA Crossed 80%, Governance Activated".
4. **User Engagement:** An email is dispatched to the Assignee containing a link to the **Virtual Agent**. The user interacts with the bot to acknowledge the risk.
5. **Data Capture:** The **UI Policy** enforces mandatory fields (SLA Breach Reason, Justification) on the Incident form once the risk flag is true.

6. **Reporting Output:** Data is aggregated into the **SLA Governance & Accountability Dashboard**, displaying widgets such as "SLA Breaches by Reason" and "Pending Justifications".

Data Dictionary Reference

Custom fields used to facilitate the data flow across the Incident table.

- **u_sla_breach_reason (Choice):** Includes options like "Dependency on third party," "System outage," and "Awaiting user response".
- **u_sla_breach_justification (String):** Captured via the Virtual Agent or Incident form.
- **u_sla_acknowledgement (True/False):** Indicates if the user has confirmed awareness of the impending breach.